

Highlights

- A 20 ka Tagua Tagua Lake record unveils earliest climate–human feedbacks in central Chile
- Multiproxy proxies reveal ecological and cultural refugia at the Pleistocene–Holocene transition
- Orbital forcing, ENSO-like, SWW and PSJ events shaped late Pleistocene hydroclimate
- Multiproxy evidence documents human–megafauna co-occurrence during abrupt hydroclimatic changes
- Early Holocene (12.5–8.6 ka BP) featured enhanced moisture and ENSO-driven pulses